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Post-quantum cryptographic authentication protocol for industrial IoT using lattice-based cryptography

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The integration of NIST-standardized lattice-based cryptographic algorithms (ML-KEM and ML-DSA) into TLS 1.3 enables post-quantum secure authentication for Industrial IoT (IIoT) environments. This work implements these algorithms within the X.509 certificate infrastructure and evaluates their performance on resource-constrained IIoT hardware (Raspberry Pi 4). Experimental measurements covering key generation, encapsulation, decapsulation, and signature operations were obtained using a liboqs-enabled TLS 1.3 stack. The results show that PQ TLS 1.3 achieves comparable handshake latency to conventional TLS on IIoT-class gateways, with certificate size identified as the dominant overhead. These findings confirm the practicality of post-quantum authentication in IIoT systems while acknowledging limitations related to the hardware scope and simulated networking. Future work will extend validation to field deployments and additional PQC candidates.

Keywords Post-Quantum Cryptography, Authentication, Industrial IoT, Quantum Threats, Lattice-Based Cryptography

The advent of quantum computing marks a pivotal shift in computational and cryptographic technologies. Large-scale quantum computers are now becoming a reality, posing threat to traditional public-key cryptography[1]. These advancements threaten the network infrastructure of every domain due to the ability of quantum algorithms Shor's to solve hard factorization problem efficiently on which all traditional public-key algorithms are based on[2].

In response to these threats, Post-Quantum Cryptography (PQC) has become active research area. On July 5, 2016, NIST initiated an open Request for Proposal invitation process, inviting researchers and mathematicians worldwide to propose new quantum-resistant public-key algorithms[3]. This process included public comments as part of the evaluation, planned to span seven to ten years, aiming to select the best PQ candidate algorithm for standardization, with a submission deadline set for November 30, 2017[4]. This multiple-round process was preparing alternate public-key key encapsulation mechanisms and digital signatures for a future where large-scale quantum computers could not break public-key algorithms and secure our current digital devices from quantum threats[5].

The National Cybersecurity Center of Excellence (NCCoE) and NIST held 'Migration to PQC Community of Interest' workshop on Aug 15 2023. The workshop provided platform for industry personnel and government sectors to connect with NIST's PQC project team, participate in panel discussions, and address challenges in the PQC transition. NCCoE also established a 'Migration to PQC' consortium, involving various collaborators to develop guidance for organizations preparing for PQC transition[6]. NCCoE also released NIST SP 1800-38A - 'Migration to PQC Preliminary Draft' which provides a guide and advisory on the necessary replacements of devices and technologies to prepare organizations against quantum computing attacks[7].

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NIST, Cybersecurity and Infrastructure Security Agency (CISA), and the National Security Agency (NSA) published a joined factsheet on Aug 17 2023 on quantum readiness which recommends a roadmap for organizations on PQC migrations, conducting risk assessments, creating cryptographic inventories and coordinating with vendors[8].

On Aug 24 2023, NIST finally released three initial drafts of algorithms selected for PQC Standard. These documents are FIPS-203 (CRYSTALS-KYBER) now ML-KEM, FIPS-204 (CRYSTALS-Dilithium) now ML-DSA, and FIPS-205 (SPHINCS+) now SLH-DSA[9]. The public comment period for these drafts lasted until November 22, 2023.

NSA also released the Commercial National Security Algorithm Suite (CNSA) 2.0 guide which provided a complete roadmap for PQC transition for organizations. NSA stated that by 2023, every public and government sector must be secured from quantum threats. This includes areas such as operating systems, web services and networking equipment. Moreover, the software and firmware signing used in the organization must be quantum-secure by 2030 as well. CNSA 2.0 recommends algorithms to be used such as, AES-256, Kyber level5, Dilithium level5, SHA 384, SHA 512, Leighton-Micali Signature (LMS) with SHA-257 or SHA-192 and the Extended Merkle Signature Scheme (XMSS)[10].

Considering these developments, our research focuses on the integration of lattice-based cryptographic protocols in the Industrial IoT environment, which focuses on migrating to Post-Quantum TLS 1.3 protocol and X.509 certificates, aiming to secure IIoT systems against quantum threats.

Amidst these developments, our research is centered on the integration of lattice-based cryptographic protocols in the IIoT landscape. We specifically delve into embedding these algorithms within Transport Layer Security (TLS) 1.3 and X.509 certificates, aiming to enhance the security and efficiency of communications over network systems. TLS 1.3, bolstered by lattice-based cryptography, is positioned as a key defense against quantum threats.

The core of our study lies in the practical implementation of specific lattice-based algorithms, such as the Ring Learning With Errors (RLWE), within TLS 1.3 and X.509 frameworks. Our objective is to showcase the algorithms' efficiency, implementation feasibility, and advanced security, highlighting their compatibility with existing infrastructures and their crucial role in countering quantum computing risks.

This research significantly contributes to the evolving landscape of PQC, offering practical insights and applications of lattice-based cryptographic methods. These efforts are particularly vital in securing IIoT systems, an area increasingly susceptible to emerging quantum computing threats.

The structure of our paper (Fig. 1) methodically explores the PQC landscape within the Industrial IoT. We commence with Section Preliminaries, reviewing existing PQC implementations. This is followed by Section IIoT security challenges and requirements, which delves into foundational concepts such as threat models and provides an overview of PQC algorithms. In Section Proposed scheme, we tackle the unique security challenges inherent in IIoT. Section Benchmarking TLS 1.3 over lattice-based cryptography is dedicated to discussing the integration of PQC into TLS 1.3 and X.509. We further extend our research in Section Performance analysis by benchmarking TLS 1.3 within a PQC framework. The paper culminates with Section Security analysis and Discussion, offering a synthesis of our findings, performance and security analysis and suggesting directions for future research in Section Conclusion . This organization ensures a comprehensive understanding of the

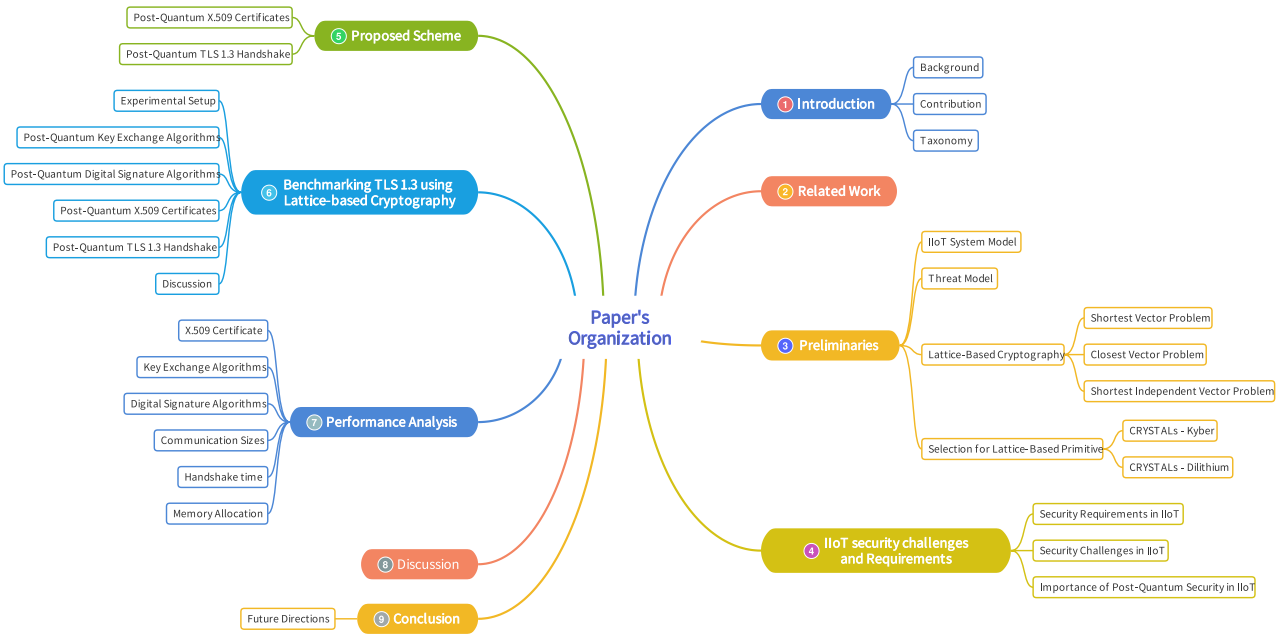

Fig. 1. Paper's Organization.

quantum computing challenge, the development of PQC, and our focused approach on deploying lattice-based cryptography in IIoT contexts.

Related work

Post-quantum authentication for the Internet of Things (IoT) and Industrial IoT (IIoT) has become a pressing research area as quantum computing threatens RSA and ECC. A dominant trend across the literature is the adoption of lattice-based algorithms such as Dilithium, Kyber, and Falcon, reflecting their selection in the NIST PQC process. Complementary schemes include hash-based SPHINCS+[12], code-based QC-LDPC[15], and multivariate or lightweight alternatives such as ANT[20].

Lohachab[29] introduced a quantum-resistant IoT framework using ECC and PQC, but it lacked a clear authentication phase and was vulnerable to replay and MITM attacks. Latif et al.[30] optimized SHA-3-based HMAC for quantum resistance, but required persistent message records. Crockett et al.[31] integrated PQC into TLS/SSH prototypes, though with outdated NIST candidates, leading to high latency. Code-based designs (e.g., Chikouche[32]) achieved confidentiality but faced replay and timing attacks. Isogeny-based authentication[33] was eliminated from NIST due to loop-abort vulnerabilities. Other works proposed lattice-based encryption for keyword search[34], hybrid PQC-TLS mechanisms[23], and vehicular privacy schemes[22], but all suffered from either overhead, non-standard assumptions, or scalability issues.

Ravi et al.[24] proposed LASAN for automotive systems using Kyber/Dilithium via pqm4, but with heavy costs. Sikeridis et al.[21] tested PQ signatures in TLS 1.3, but used outdated round-2 candidates. Paul et al.[25] explored mixed PQ/traditional certificate chains for TLS 1.3, easing transition but limiting security levels. More recently, McLoughlin et al.[17] prototyped PQ-DTLS 1.3 on Raspberry Pi 4, showing feasibility but with $6\times$ higher packet overhead and 800 KiB extra memory for 100 devices.

Behnia and Yavuz[20] introduced the ANT signature, achieving 8–16$\times$ faster signing than Dilithium on 8-bit AVR nodes, with 5.5$\times$ smaller payloads. Banegas et al.[19] benchmarked LMS, Dilithium, and Falcon on Cortex-M and RISC-V, highlighting trade-offs in verification latency vs. key sizes. Schöffel et al.[18] tested Kyber, SABER, NTRU, and SIKE on Cortex-M4, finding 2.5–4 KB memory overhead and bandwidth demands surpassing ECC. Gupta[16] proposed identity-based lattice authentication to minimize certificate load, though without hardware validation.

Samandari and Gritti[12] tested Dilithium, Falcon, and SPHINCS+ in a three-layer smart-agriculture setting, showing 45% reduced communication overhead when offloading to gateways. In a follow-up[13], they compared signatures vs. KEMs in MQTT, finding KEMs 71% faster but with message sizes up to 17 KB. Kumari et al.[15] proposed a lattice/code hybrid reducing authentication delay by 23%, validated via FPGA. Liu et al.[28] presented a lattice-based multi-party collaborative protocol, resistant to replay and desynchronization attacks.

Prajwal et al.[14] combined qPUFs, XOR, and hashing for IIoT authentication on NodeMCU, formally verified with Scyther. Tawfeeq et al.[11] applied Dilithium in drone networks for mission-critical authentication. Other sectoral adaptations include fog computing[26], vehicular communication[22], and industrial OPC-UA protocols[23].

Overall, PQC-based authentication is feasible for IoT/IIoT, but challenges persist: (i) signature sizes and certificate overhead remain high, (ii) latency and memory demands strain resource-constrained devices, (iii) security comparison is inconsistent [our work] vs. Scyther[14]), and (iv) many proposals lack real-world or large-scale evaluation. While hierarchical or hybrid designs reduce latency by 23% and communication by 45%, they often introduce new complexities in scalability and certificate management. Our work advances this line by combining Dilithium and Kyber in a server-assisted, CA-based four-phase protocol, and experimentally deployed on Raspberry Pi, addressing both security rigor and IIoT-specific resource constraints (see table. 1).

Preliminaries

IIoT system model

We adopt a three-tier IIoT architecture with (i) *edge devices* (sensors/actuators/controllers, often MCU-class and battery powered), (ii) *platform/edge gateways* (Linux/SBC nodes aggregating field buses to IP), and (iii) *enterprise/back-end* services (data/PKI/management) (Fig. 2).

Data move upward with increasing aggregation and latency; control may flow downward with real-time constraints. Authentication can terminate at gateways (common in brownfield deployments) or at servers, while devices initiate client authentication as policy dictates. Links are lossy and jittery; clocks on controllers may drift by tens of milliseconds.

Threat model

We consider active network adversaries with harvest-now-decrypt-later intent and local attackers with physical proximity to edge devices.

Capabilities. Packet injection/replay/downgrade; transcript observation; compromise of outdated classical algorithms; local timing/power side-channels and limited fault injection on edge MCUs; software supply-chain tampering of firmware images.

Assets. Session keys; device identities (certificates/keys); firmware; control commands; telemetry.

Attack surfaces. (1) *Network*: replay/MiTM/downgrade; (2) *Endpoint*: side-channels, key exfiltration from insecure NVM; (3) *Lifecycle*: rogue/expired certificates, clock skew undermining freshness, malicious update paths.

Model. Analysis follows a CK-style authenticated key-exchange view with explicit entity authentication and channel key indistinguishability. KEMs are assumed IND-CCA2 and signatures EUF-CMA secure under their standard lattice assumptions; lossy links and retransmissions are accounted for.

Ref.	Purpose	Technique	Limitations
11	Mission-critical authentication for Internet of Drones (IoD)	Dilithium-based signatures integrated into flight control/auth	● Limited implementation details ● No hardware profile/scalability study ● Air-to-ground latency not evaluated
12	Three-layer IIoT auth offloading in smart agriculture	Gateway offloading; eval Dilithium/Falcon/SPHINCS+; hierarchical design	● Gateway complexity/cert management ● No large-scale field deployment
13	Compare signature vs KEM-based auth over MQTT	MQTT auth; KEMs faster (71%) but larger messages (up to 17 KB)	● Bandwidth overhead at brokers ● No stress tests under lossy links
14	qPUF-aided IIoT authentication on NodeMCU	qPUFs + XOR + hashing; Scyther-based validation	● qPUF practicality/availability concerns ● Not aligned with NIST PQ standard primitives
15	Hybrid lattice/code for smart-home auth	LWE + QC-LDPC; FPGA validation; 23% delay reduction	● Complexity of hybrid design ● Large public parameters; deployment fit
16	Reduce certificate overhead via identity-based lattice AKE	IBE-style lattice AKE to minimize PKI header	● No hardware evaluation ● PKG trust/escrow implications
17	Feasibility of PQ-DTLS 1.3 on Raspberry Pi 4	DTLS 1.3 with PQ auth; prototype on RPi4	● ~6× packet overhead; ~800 KiB for 100 devices ● Workload not IIoT-specific
18	Benchmark PQ KEM/DSA on Cortex-M4	Kyber, SABER, NTRU, SIKE; memory/bandwidth profiling	● Higher RAM (2.5–4 KB) and bandwidth than ECC ● Some eliminated candidates
19	Secure update pipelines on MCUs	LMS, Dilithium, Falcon on Cortex-M/RISC-V; benchmarks	● No end-to-end auth framework ● Trade-offs: key size vs verify latency
20	Lightweight signatures for constrained nodes	ANT signature: 8–16× faster sign; 5.5× smaller payloads on 8-bit AVR	● Non-standard; security assumptions underexplored ● Side-channel resistance not analyzed
21	PQ Signature Algorithms certificate Study for hybrid authentication	OQS-OpenSSL Libraries used on NIST's 2nd round signature algorithms	● Traditional KEM ● 2nd Round's outdated ● and eliminated candidates
22	Conditional privacy-preserving authentication in vehicular communication using lattices	NTRU-251 based on ISI and ISIS problem	● Partial key exposure attacks ● Heavy computation cost ● Non – Standard PQC
23	Proposed PQC hybrid KEM for the industrial communication protocols	Hybrid handshake with RSA-2048 + NewHope512/NTRU/Kyber90s/ LightSaber	● Required more resources ● Not feasible for IoT devices ● Non-standard algorithms
24	Proposed lattice-based secure authentication for Automotive	LASAN protocol based on Kyber's and Dilithum's pqm4 library for ECUs authentication on LWE/LWR problem	● Heavy computation ● High communication cost ● Outdated Library (2nd Round)
25	Proposed Mixed certificate chains for PQ Authentication	NIST's 3rd Round candidates + traditional ECDSA used for mix certificates chain	● Hybrid with traditional ECDSA ● Includes Non-Standard algorithms ● Only NIST security level 2 variants
26	Proposed lattice-based multi-blocks and mixed incremental signature protocol for Fog computing	Lattice-based incremental hash function scheme on SIS problem	● Desynchronization attack ● PQ Algorithm not mentioned
27	Proposed lattice-based PEAKS scheme for multi-user settings for IIoT environment	Lattice-based trapdoor scheme under random oracle model achieving IND-CKA	● Vulnerable to; Impersonation, ● Replay, side-channel attacks ● PQ Algorithm not mentioned
28	Proposed quantum secure multi-party authentication for IIoT environment	Multigeniture authentication using DAPS with Aigis and Dilithium	● Replay attack ● Desynchronization attack ● Non-Standard Signature Algorithm

Table 1. Comprehensive Literature Review on Post-Quantum Authentication including IoT and IIoT Settings.

Security goals

The Industrial Internet of Things (IIoT) demands security mechanisms that ensure the reliability, confidentiality, and integrity of data across heterogeneous and resource-constrained environments. The key security goals can be summarized as follows:

1. **Mutual authentication:** Ensure that both the device and the control server can verify each other's identities to prevent unauthorized access or impersonation.
2. **Forward secrecy:** Session keys must remain secure even if long-term device credentials are later compromised, preserving confidentiality over time.
3. **Replay and downgrade resistance:** Communication protocols should prevent attackers from reusing captured messages or forcing negotiation to weaker cryptographic parameters.
4. **Efficiency on constrained nodes:** Security operations must remain lightweight to meet IIoT devices' limitations in memory, processing power, and energy consumption.
5. **Side-channel resilience:** Implementations should minimize information leakage through timing, power, or electromagnetic emissions, ensuring robustness against physical and microarchitectural attacks.

Post-quantum cryptography

NIST's PQC effort has selected lattice-based ML-KEM (CRYSTALS-Kyber) for key establishment and lattice-based ML-DSA (CRYSTALS-Dilithium) and Falcon, plus hash-based SPHINCS+, for signatures[3,35]. We use the term "Kyber" for ML-KEM and "Dilithium" for ML-DSA.

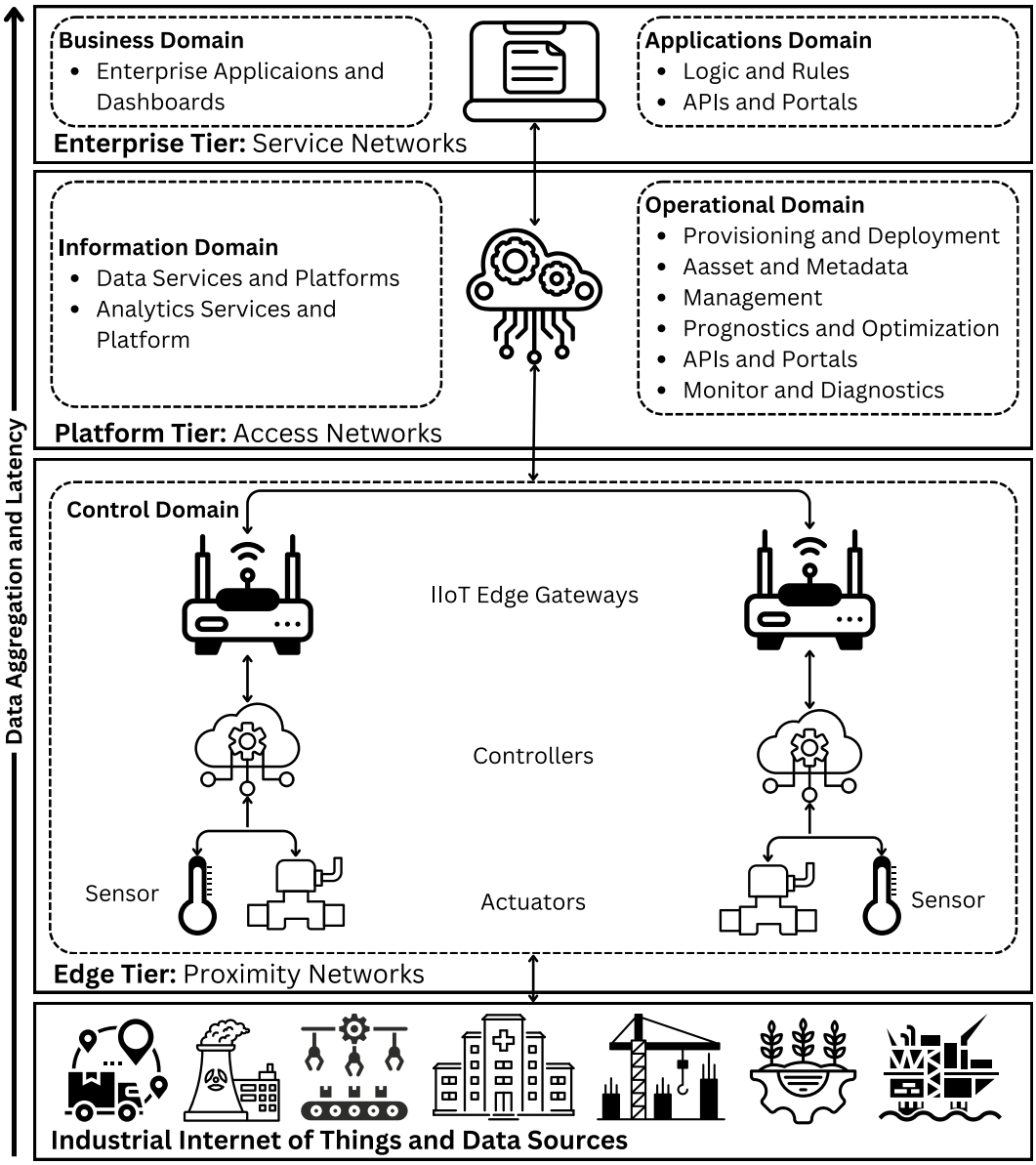

Fig. 2. IIoT 3-Tier System Model (authors' illustration).

Post-quantum primitive classes
The principal families relevant to IIoT (shown in Fig. 3): *lattice-based* (LWE/MLWE, NTRU; efficient and broadly implemented), *code-based* (e.g., McEliece; fast but large keys), and *hash-based* (e.g., SPHINCS+; conservative, larger signatures). We focus on lattice primitives for their performance/size balance on embedded targets.

Lattice-based cryptography

Lattice security rests on the hardness of classic problems such as the *Shortest Vector Problem* (SVP) and *Closest Vector Problem* (CVP), and their module/ring instantiations used by Kyber/Dilithium. We keep brief, IIoT-relevant summaries here and move tutorials to an appendix to streamline the main text.

Shortest Vector Problem (SVP)
The Shortest Vector Problem (SVP) is one of the foundational problems in lattice theory and has direct implications for lattice-based cryptography. A lattice $\mathcal{L}(\mathrm{B})$ in $\mathbb{R}^n$ is the set of all integer linear combinations of a basis $\mathrm{B} = (b_1, b_2, \ldots, b_k)$. The SVP requires finding the nonzero lattice vector of minimal Euclidean norm, i.e., the shortest nonzero vector in the lattice.

Formally, given $\mathcal{L}(\mathrm{B})$, the task is to find a nonzero Bx with $\mathrm{x} \in \mathbb{Z}^k$ such that:

$$\|\mathrm{Bx}\| \leq \gamma \cdot \lambda_1(\mathcal{L}),$$

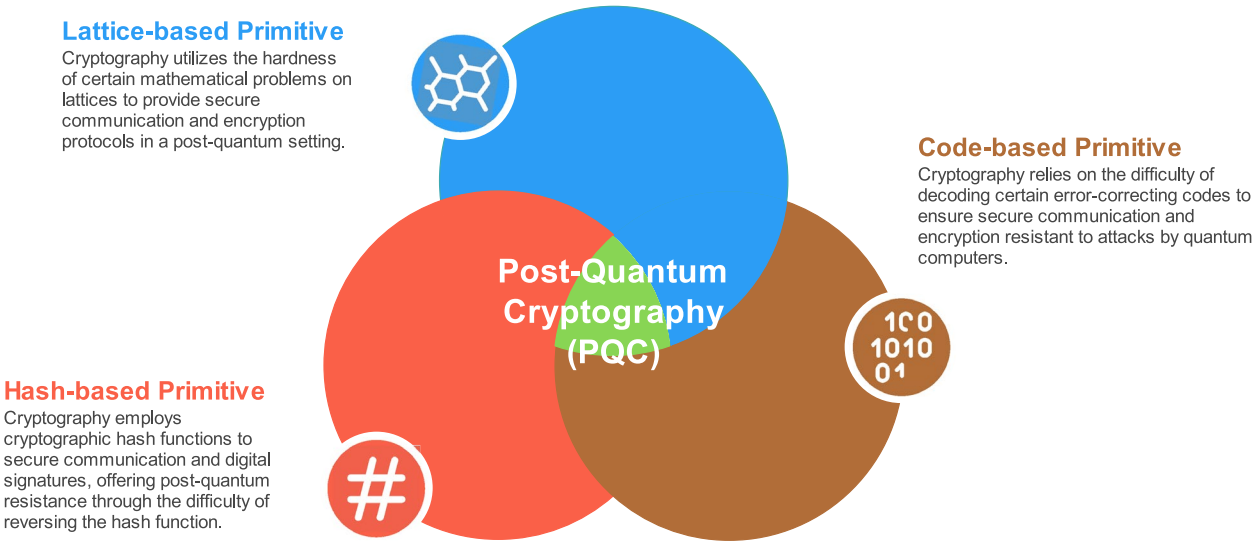

Fig. 3. Post-quantum primitive families.

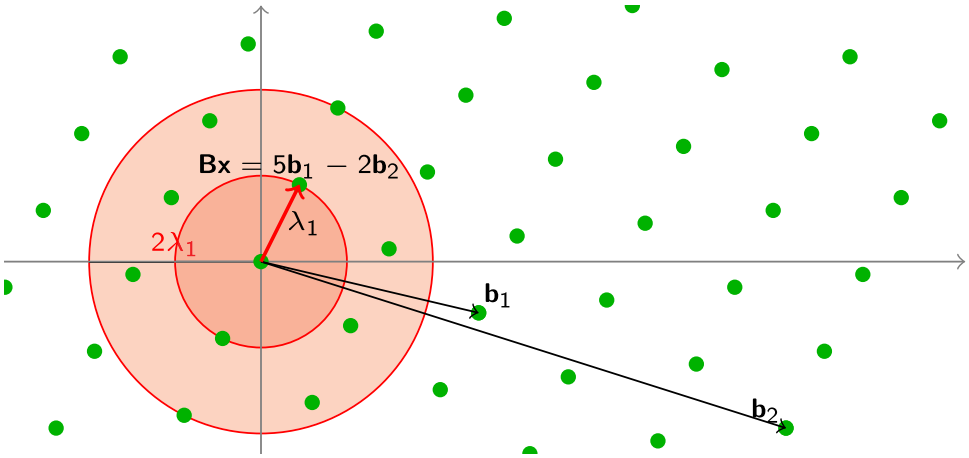

Fig. 4. Shortest Vector Problem (SVP). The shortest vector may be a nontrivial linear combination, e.g., $5b_1 - 2b_2$.

where $\lambda_1(\mathcal{L})$ denotes the first successive minimum of the lattice and $\gamma \geq 1$ is an approximation factor. In the exact SVP, $\gamma = 1$.

SVP is known to be NP-hard under randomized reductions and is therefore widely regarded as computationally intractable. This hardness underpins the security of lattice-based cryptographic constructions. Approximate versions (γ-SVP) are particularly relevant for cryptography, as they capture the difficulty of finding vectors that are "short enough" rather than optimally short.

The challenge of SVP becomes apparent when the given basis vectors are long or nearly collinear. In such cases, much shorter vectors exist as integer combinations of the basis (e.g., $5b_1 - 2b_2$), but identifying them is computationally hard. One can visualize the process by expanding a sphere centered at the origin until it first intersects a nonzero lattice point; this first point represents the shortest vector (see Fig. 4).

Algorithms such as LLL (Lenstra–Lenstra–Lovász) reduction and its stronger variants (BKZ, BKZ-2.0) can find relatively short vectors, but they cannot solve SVP exactly in high dimensions. Even approximate solutions remain difficult beyond modest approximation factors. This intractability is precisely what makes SVP a cornerstone assumption in post-quantum cryptography: its presumed hardness ensures security against both classical and quantum adversaries.

Closest Vector Problem (CVP)

The Closest Vector Problem (CVP) generalizes SVP and is equally important in lattice theory and cryptography. Given a lattice $\mathcal{L}(\mathrm{B})$ and a target vector $\mathrm{t} \in \mathbb{R}^n$, the goal is to identify the lattice vector Bx closest to t:

$$\|\mathrm{Bx} - \mathrm{t}\| \leq \gamma \cdot \mu(\mathcal{L}, \mathrm{t}),$$

where $\mu(\mathcal{L}, \mathrm{t})$ denotes the distance from t to the lattice, and γ is an approximation factor.

Visually, CVP asks: given a lattice defined by basis vectors b_1, b_2 and a target t, which lattice point lies nearest to t? Whereas SVP seeks the shortest vector from the origin, CVP seeks the closest vector to an arbitrary target point 5.

CVP is at least as hard as SVP and is often more difficult. Approximating CVP within certain factors is NP-hard, and no efficient algorithms are known in general. Approximate versions (Approx-CVP) are useful in practice, particularly in cryptography. For example, in Learning With Errors (LWE)-based encryption, decryption essentially reduces to an instance of CVP: given a noisy target vector, recover the closest lattice point that corresponds to the original message. Thus, the presumed hardness of CVP underpins the security of widely deployed PQC candidates such as Kyber and FrodoKEM.

Shortest Independent Vector Problem (SIVP)

The Shortest Independent Vector Problem (SIVP) extends the hardness of SVP to multiple dimensions. Instead of finding a single short vector, SIVP requires finding *n* linearly independent short vectors, where *n* is the rank (dimension) of the lattice.

Formally, given a lattice $\mathcal{L}(\mathrm{B})$, the goal is to find linearly independent vectors $\mathrm{Bx}_1, \mathrm{Bx}_2, \ldots, \mathrm{Bx}_n \in \mathcal{L}$ such that:

$$\max_i \|\mathrm{Bx}_i\| \leq \gamma \cdot \lambda_n(\mathcal{L}),$$

where $\lambda_n(\mathcal{L})$ denotes the *n*-th successive minimum of the lattice.

Conceptually, one can think of expanding a sphere around the origin until it encloses *n* linearly independent lattice vectors. These vectors collectively approximate the geometry of the lattice at the *n*-th successive minimum λ_n. Unlike SVP, which focuses only on λ_1, SIVP requires a full "short basis" of the lattice.

As with SVP and CVP, approximate formulations of SIVP are typically considered, where the requirement is to find vectors not much longer than λ_n. Approximate SIVP remains hard in high dimensions and is a central assumption in lattice-based cryptography. Indeed, worst-case to average-case reductions show that solving approximate SIVP is as hard as breaking Learning With Errors (LWE), thereby grounding the security of many lattice-based primitives 6.

In summary, SVP, CVP, and SIVP collectively form the hardness foundation of lattice-based cryptography. Their computational intractability ensures that schemes based on these problems, such as NTRU, Kyber, and Dilithium, remain secure even against quantum adversaries. Together, they provide the theoretical guarantees that make lattice-based cryptography one of the most promising families of post-quantum secure schemes.

Why lattices for IIoT

Lattice primitives provide (i) *post-quantum assurance* with widely analyzed assumptions (LWE/MLWE, NTRU), (ii) *performance* suitable for embedded targets, and (iii) *standards alignment* with NIST's selections. We avoid tutorial exposition and instead provide actionable guidance below.

Selected PQC algorithms

ML-KEM (CRYSTALS-Kyber) Module-LWE KEM with IND-CCA2 security via FO transform; commonly profiled at levels roughly corresponding to Cat. I/III/V (Kyber512/768/1024)[36]. We use Kyber for handshakes and derive application keys via HKDF. Constant-time, low-RAM implementations exist for MCU/SBC platforms (see table. 2).

ML-DSA (CRYSTALS-Dilithium) Module-SIS/LWE signature with EUF-CMA security; recommended at levels roughly mapping to Cat. II/III/V (Dilithium2/3/5)[37]. We adopt constant-time sampling and 12-round Keccak-based expanders where appropriate (see table. 3).

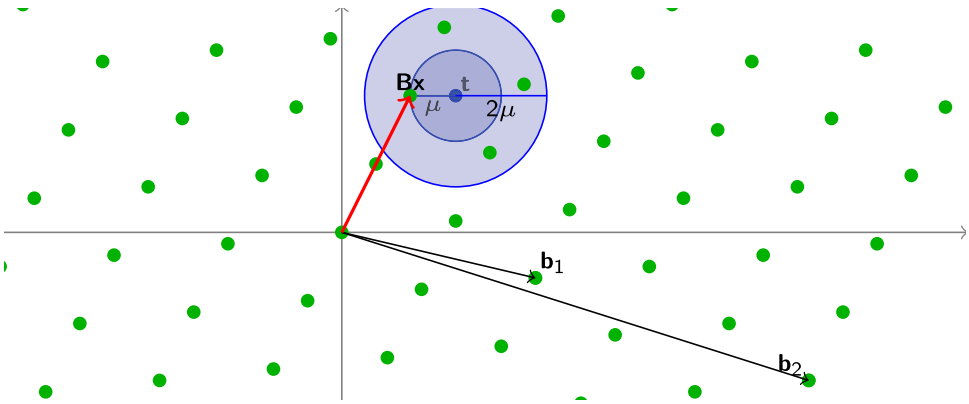

Fig. 5. Closest Vector Problem (CVP). The goal is to find the lattice point closest to target t.

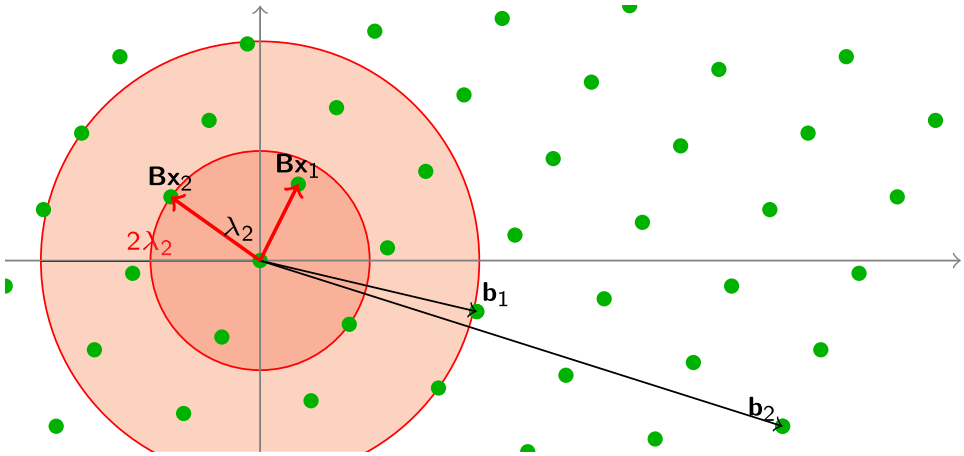

Fig. 6. Shortest Independent Vector Problem (SIVP). The task is to find n short, linearly independent vectors.

KEM	Security (indicative)	Public Key (bytes)	Ciphertext(bytes)
Kyber512	Cat. I	800	768–800
Kyber768	Cat. III	1184	1088
Kyber1024	Cat. V	1568	1568

Table 2. Indicative sizes for Kyber parameter sets.

DSA	Security (indicative)	Public Key (bytes)	Signature (bytes)
Dilithium2	Cat. II	1312	2420
Dilithium3	Cat. III	1952	3293
Dilithium5	Cat. V	2592	4595

Table 3. Indicative sizes for Dilithium parameter sets.

Tier	KEM	Signature
Sensor (MCU)	Kyber512	Dilithium2
Edge (SBC)	Kyber768	Dilithium3
Gateway/Server	Kyber1024	Dilithium5

Table 4. Recommended PQ suites by device tier (deployment profiles).

Parameterization and device profiles for IIoT

To connect primitives to hardware constraints, we define pragmatic profiles (table. 4):

Rationale. Sensors prioritize smaller ciphertexts and faster decapsulation; edges trade modest overheads for higher assurance; gateways/servers anchor PKI and can absorb Cat. V costs. These choices bound handshake bytes and peak memory while retaining clear migration paths.

IIoT security challenges and requirements

Industrial IoT (IIoT) couples long-lived, heterogeneous devices with real-time control. Security controls therefore must (i) authenticate endpoints under lossy links and clock drift, (ii) bound latency and memory on constrained nodes, and (iii) remain secure against harvest-now-decrypt-later quantum adversaries. Unlike generic IoT surveys, we link each challenge to a *post-quantum* mechanism and to the concrete *design choices* used in our protocol (see Fig. 7).

Requirements mapped to PQC and Our scheme

We employ lattice-based ML-KEM (Kyber) for key establishment and ML-DSA (Dilithium) for authentication (see 5). Authentication terminates at gateways or servers depending on the device tier; sensors can perform client authentication when mandated by policy.

Fig. 7. IIoT challenges and requirements (overview).

Challenge	PQC relevance (mechanism)	Our scheme choice (why)	Impact in IIoT
Low-power sensors (MCU, battery)	Handshake and signatures must fit tight RAM and energy budgets. Use smallest practical KEM/DSA.	Kyber512 + Dilithium2 at sensors Parameterization and device profiles for IIoT; offload certificate path checks to gateway when policy permits.	Reduces on-device bytes and cycles; bounded peak RAM; minimizes retransmissions on lossy links.
Edge gateways (SBC, Linux)	Can sustain stronger parameters while aggregating many sessions.	Kyber768 + Dilithium3 at edges Parameterization and device profiles for IIoT to raise assurance without large latency penalty.	Balances throughput and security for broker/MQTT/TLS concentrators.
High-throughput/back-end	Anchors PKI and terminates many handshakes; prioritizes strongest security.	Kyber1024 + Dilithium5 at gateways/servers for anchoring trust.	Future-proofs against long-term attacks; negligible impact on server-class hardware.
Harvest-now-decrypt-later	Classical DH/ECDH handshakes are vulnerable to future decryption.	All sessions use Kyber KEM for key establishment; hybrid optional if policy requires.	Confidentiality of captured traffic preserved post-quantum.
Replay/downgrade on lossy links	Negotiation can be forced to weaker suites; retransmissions can enable replay.	TLS 1.3 with PQ suites only; strict signature-algorithm/KEM allow-lists; transcript-bound keys.	Prevents downgrade; nonces + transcript hashing block replay under packet loss.
Side-channels on edge MCUs	Secret-dependent timing in PQC can leak keys.	Constant-time Kyber/Dilithium implementations; no secret-dependent branches; RNG health checks.	Mitigates timing/power leakage; aligns with Security Goal 5 Security goals.
Certificate size/PKI overhead	PQ signatures/keys enlarge X.509 objects and handshake bytes.	Short-lived device certs; compressed chains; gateway-assisted validation; Dilithium2 at the edge where feasible.	Keeps handshake bytes B manageable; reduces revalidation load on sensors.
Firmware/auth update integrity	Long device lifetimes require quantum-safe signing of artifacts.	Dilithium signing for images; verification on device/gateway; secure boot roots on gateways.	Prevents supply-chain tampering across lifecycle.
Real-time and jitter sensitivity	Control loops require capped handshake latency T_{hs}.	Tiered parameters Parameterization and device profiles for IIoT; session resumption and PSK for steady-state.	Meets latency budgets; avoids frequent full PQ handshakes in tight loops.

Table 5. IIoT challenges mapped to PQC mechanisms and our deployment choices.

Security requirements

Authentication. Mutual authentication uses Dilithium certificates (server-side mandatory; client-side policy-driven). Profiles in Parameterization and device profiles for IIoT select Dilithium2/3/5 by tier; this directly answers low-power vs. high-assurance deployments.

Key establishment and forward secrecy. All handshakes employ Kyber (ML-KEM). Optional hybrid modes are supported for regulatory migration but disabled by default to avoid downgrade complexity.

Access control. Authorization is handled at the application layer, but credential binding anchors to PQ X.509 identities (reduces spoofing when classical roots are deprecated).

Data security. Application keys are derived from PQ handshakes (HKDF) and used with AEAD; confidentiality remains safe against HNDL threats.

Maintainability/resilience. PQ-signed firmware and config updates; session resumption to cap latency; certificate rotation policies tuned to device class.

Monitoring. PQ does not replace IDS/telemetry; however, handshake metadata (suite, sizes) enables policy checks to detect illicit non-PQ sessions.

IIoT-specific challenges

Long-lived components. Use PQ roots and Dilithium-signed updates; define migration paths for legacy nodes via gateway termination.

High connectivity / IT–OT convergence. Enforce PQ-only TLS on northbound links; use Kyber/Dilithium profiles per tier to maintain throughput on aggregation points.

Device proliferation. Automate enrollment with PQ certificate templates; prefer gateway-assisted chain validation to keep MCU costs low.

Human error / sabotage. Short-lived credentials, revocation via gateway CRLs/OCSP; secure default suites (no classical fallbacks).

Why post-quantum for IIoT

Classical ECDH/ECDSA handshakes jeopardize confidentiality under harvest-now-decrypt-later. Lattice-based Kyber/Dilithium provide efficient PQ security on embedded and edge platforms, and–when parameterized per device tier–respect IIoT latency and memory constraints. Our server-assisted design (gateways/servers terminate heavy validation; sensors use lighter profiles) connects PQC directly to the operational realities of IIoT, avoiding generic "checklists" while meeting the goals in Security goals.

Proposed scheme

Our contribution is a migration strategy that embeds lattice-based post-quantum algorithms into the full TLS 1.3 stack for IIoT environments. Unlike prior survey-style or purely theoretical works, we (i) integrate ML-DSA (CRYSTALS-Dilithium) into X.509 certificates for authentication, (ii) adopt ML-KEM (CRYSTALS-Kyber) for handshake key exchange, and (iii) benchmark the complete protocol on IIoT-class hardware (Raspberry Pi 4). This combination–practical TLS 1.3 migration, certificate integration, and performance/resource evaluation–addresses the unique requirements of IIoT, where devices are resource-constrained, expected to operate for decades, and must resist harvest-now-decrypt-later threats.

Post-quantum X.509 certificates

A central aspect of our scheme is adapting X.509 certificates to lattice-based signatures. Figure 8 illustrates the structure of a PQ X.509 certificate embedding Dilithium public keys and signatures. Compared with classical ECDSA/RSA certificates, sizes increase by 1.5–2.5×, but remain manageable when offloaded to gateways. For constrained clients, validation of short-lived end-entity certificates can be deferred to gateways or performed incrementally.

In our implementation:

- The *Root CA* uses Dilithium-5 to anchor long-term trust.
- Intermediate and end-entity certificates employ Dilithium-2/3, balancing security with device verification costs.

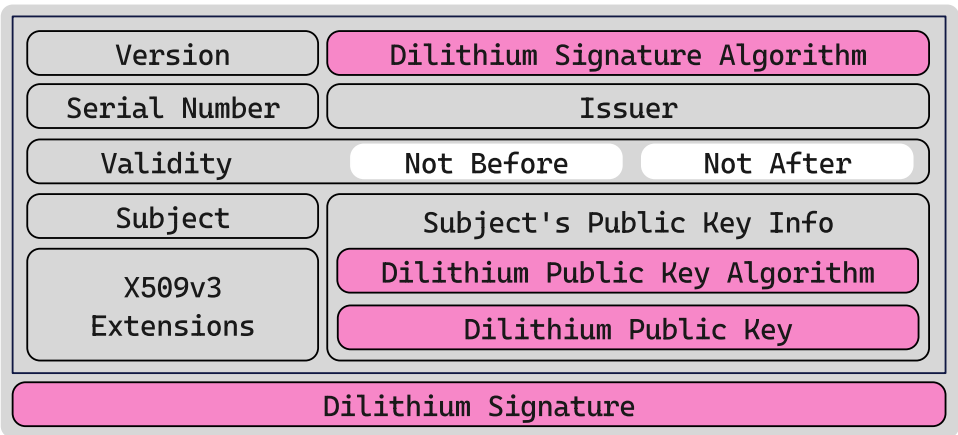

Fig. 8. Lattice-based X.509 certificate with Dilithium signatures.

- Certificate chains are compressed and rotated frequently, reducing the overhead on sensors.

This design preserves PKI compatibility while enabling IIoT deployments to transition without abandoning existing trust infrastructures.

Post-quantum TLS 1.3 handshake

We extend the TLS 1.3 handshake (Fig. 9) by replacing classical primitives with lattice-based counterparts:

- **Key establishment:** The ephemeral Diffie–Hellman exchange is replaced by Kyber encapsulation/decapsulation (Kyber512 for MCUs, Kyber768 for gateways, Kyber1024 for servers).
- **Authentication:** CertificateVerify and Finished messages are signed with Dilithium, ensuring EUF-CMA security even against quantum adversaries.
- **Optional client authentication:** When required by policy (e.g., actuator-to-gateway links), clients authenticate using Dilithium-2.

The handshake flow mirrors classical TLS 1.3, but with larger key and signature objects. Our evaluation 9 quantifies these costs: on Raspberry Pi 4, handshake completion time increased by ∼1.6× relative to ECDHE+ECDSA, with RAM consumption rising by ∼600 KiB. However, these overheads remain acceptable for gateways and servers, while sensors benefit from shorter parameter sets and gateway-assisted validation.

Why IIoT needs this scheme

IIoT deployments differ from enterprise IT in three key ways:

1. **Resource-constrained nodes.** Kyber512 and Dilithium2 provide quantum resistance while fitting MCU-class limits.
2. **Long-lived devices.** Equipment expected to operate for 10–20 years must resist harvest-now-decrypt-later attacks; PQC ensures confidentiality of today's traffic remains intact against future quantum adversaries.
3. **Gateway-centric trust.** Offloading certificate chain validation and large signature operations to gateways enables constrained sensors to participate securely without excessive overhead.

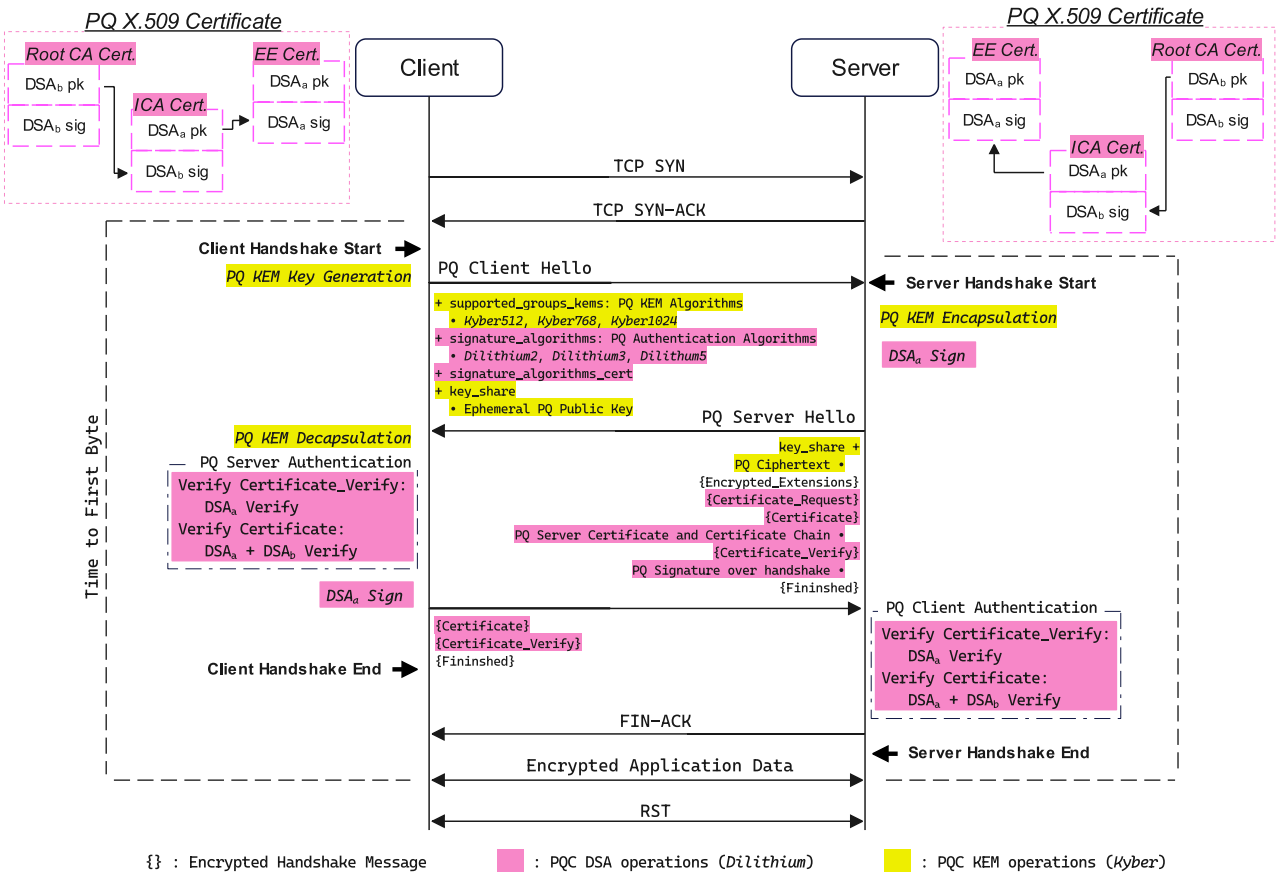

Fig. 9. TLS 1.3 handshake adapted with Kyber KEM and Dilithium signatures.

By combining TLS 1.3 migration with lattice-based primitives and demonstrating feasibility on real IIoT-class hardware, our scheme offers a practical, forward-compatible path that balances quantum security, interoperability, and resource constraints.

Benchmarking TLS 1.3 over lattice-based cryptography

We benchmark a complete post-quantum (PQ) TLS 1.3 stack tailored to IIoT constraints, focusing on (i) microbenchmarks of KEMs and DSAs, (ii) PQ-ready X.509 sizes, and (iii) end-to-end TLS 1.3 handshake costs. Our goal is not to re-establish known desktop/server results, but to characterize performance on IIoT-class hardware and provide deployment guidance by device tier.

We build on prior TLS/PQC evaluations: Sikeridis et al. (Intel CPUs/cloud)[21], Tasopoulos *et al.* (STM32 microcontrollers; performance/energy)[38,39], and Paul *et al.* (mixed certificate chains across notebook/cloud)[25]. Our results complement these by targeting a common IIoT gateway platform (Raspberry Pi 4) and by reporting end-to-end handshake/footprint with Kyber/Dilithium and PQ X.509.

Experimental setup

Platform. Raspberry Pi 4 Model B (Broadcom BCM2711, quad-core Cortex-A72 @ 1.8 GHz, 4 GB LPDDR4-3200), Ubuntu 23.04. The device plays both client and server roles to isolate cryptographic costs.

Software. PQ algorithms from `liboqs 0.8.0`; TLS from `wolfSSL 5.6.3` (OpenSSL-compatible fork) integrated with `liboqs` for PQ TLS 1.3 and PQ X.509 generation. Cipher suite: `TLS_AES_256_GCM_SHA384`. We evaluate Kyber (512/768/1024) and Dilithium (2/3/5); selected classical baselines are included for reference.

Methodology. Client/server processes run on the same host (loopback) to eliminate variable WAN/LAN latency and isolate cryptographic and framing overheads. Unless stated, averages are over repeated runs with warm caches.

Limitations. Results reflect one IIoT-class platform, local loopback (no network delay/jitter), and one TLS implementation. This isolates crypto and certificate handling but under-represents real deployments with hops, loss, and congestion Threats to validity and limitations.

Benchmark repeatability and metrics.
Each benchmark was executed 30 consecutive times on the Raspberry Pi 4 testbed under identical thermal and CPU-governor conditions (performance mode, 1.8 GHz). The mean and standard deviation were computed for all timing metrics. Handshake times and key/verify operations are reported in milliseconds (ms), while key and ciphertext sizes are in bytes (B). All tests were performed on a local loopback network to exclude variable network latency; therefore, results reflect cryptographic computation only.

KEM microbenchmarks (key establishment)

Table 6 reports sizes and operation timings for Kyber (512/768/1024) and classical ECDH/FFDHE baselines (OpenSSL 3.0.8). Despite larger objects, Kyber shows low keygen/encap/decap latency on RPi4. Figure 11 contrasts public/secret/ciphertext sizes; Fig. 10 (log scale) shows Kyber's consistently low operation times relative to classical key agreement. These confirm that PQ KEM cost is compatible with gateway-class IIoT hardware, while giving the confidentiality benefits against harvest-now-decrypt-later.

DSA microbenchmarks (authentication)

Table 7 presents Dilithium (2/3/5), Falcon (512/1024), and classical ECDSA/RSA. Figures 13 and 12 summarize size and timing trade-offs. Dilithium offers strong verification throughput with moderate signing costs, making it suitable for server/gateway authentication and, where policy requires, client authentication on capable nodes.

PQ X.509 object sizes

Table 8 and Fig. 14 quantify key/CSR/certificate sizes for Dilithium vs. RSA. PQ certificates are 1.5–6× larger than ECDSA/RSA, reinforcing the need for gateway-assisted validation and short chains in sensor-facing deployments. This overhead is the principal contributor to increased handshake bytes (next subsection).

					Key gen		En-caps		De-caps	
KEM Algorithm	Hardness Problem	Public key	Secret key	Cipher text	Mean	ops/s	Mean	ops/s	Mean	ops/s
Kyber-512	Module Learning With Errors	800	1632	768	0.021	47619.050	0.028	35714.290	0.020	50000.000
Kyber-768		1183	2400	1088	0.034	29411.760	0.041	24390.240	0.030	33333.330
Kyber-1024		1568	3168	1568	0.046	21739.130	0.056	17857.140	0.042	23809.520
P-256[1]	Elliptic Curve Discrete Logarithm	32	32	70	12.711	78.670	0.100[4]	7711.500[4]	-	-
P-384[2]		48	48	103	18.082	55.300	1.500[4]	673.000[4]	-	-
P-521[3]		66	66	138	14.581	68.580	0.700[4]	1444.400[4]	-	-
FFDHE 3072	Finite Field Discrete Log	256	256	256	32.100	31.090	1.800[4]	554.900[4]	-	-

Table 6. Key Exchange Algorithms. [1]ECDH 256-bits (nistp256), [2]ECDH 384-bits (nistp384), [3]ECDH 521-bits (nistp521), [4]Key agreement, Units: Key sizes/ciphertext (bytes), Mean (ms).

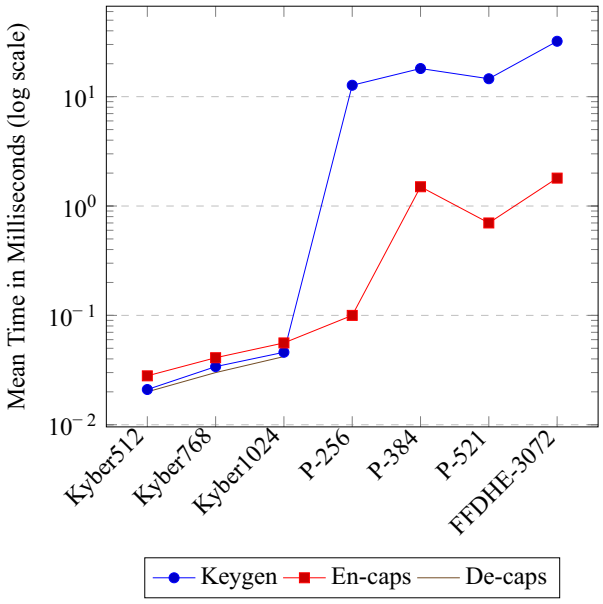

Fig. 10. Traditional vs. PQ KEX: Keygen, En-caps & De-cap ops.

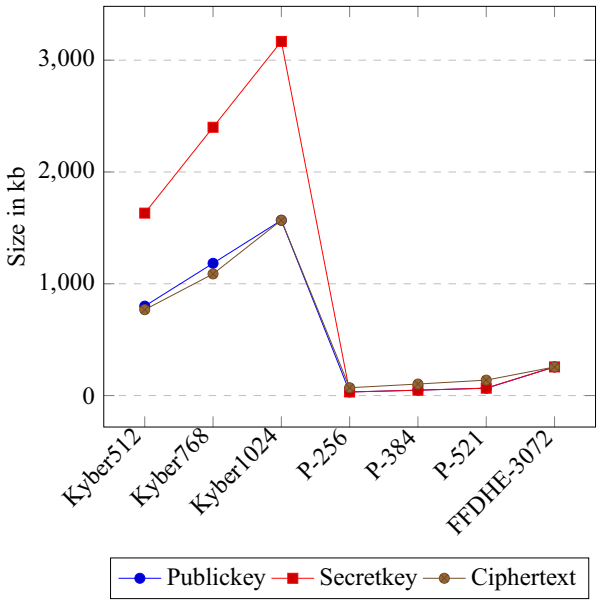

Fig. 11. "Traditional vs. PQ KEX: Publickey, Secretkey & Ciphetext Sizes.

End-to-End PQ TLS 1.3 handshake

We instrument end-to-end handshakes under combinations of Dilithium (auth) and Kyber (KEM), plus a SECP256R1 baseline. Figure 15 shows a representative Wireshark trace.

Table 9 reports client/server heap, stack, total communication (bytes), and handshake time (ms) per combination. Figures 19, 17 summarize communication and latency and fig. compares stack allocation. Key observations:

- **Communication.** PQ handshakes add tens of kilobytes over classical due to PQ X.509 sizes; nevertheless, in our setup several Kyber/Dilithium pairs transmitted *less* than some classical baselines with larger certificate chains.
- **Latency.** Client handshake times for Kyber512/Dilithium2–3 are typically 6.5–8.0 ms; Kyber1024/Dilithium5 rise toward 12–13 ms. The SECP256R1 baseline spans ∼12–20 ms in our configuration, illustrating that PQ does not necessarily imply higher latency on this platform.

Signature Algorithm	Hardness Problem	Public key	Secret key	Signature	Key gen		Sign		Verify	
					Mean	ops/s	Mean	ops/s	Mean	ops/s
Dilithium2	Module Learning with Errors	1312	2528	2420	0.064	15625.000	0.171	5847.950	0.064	15625.000
Dilithium3		1952	4000	3293	0.107	9345.790	0.280	3571.430	0.104	9615.380
Dilithium5		2592	4864	4595	0.168	5952.380	0.334	2994.010	0.164	6097.560
Falcon512	NTRU	897	1281	666	14.642	68.280	0.495	2020.200	0.091	10989.010
Falcon1024		1793	2305	1280	44.012	22.720	0.975	1025.640	0.177	5649.720
P-256[1]	Elliptic Curve Discrete Logarithm	32	32	70	14.775	67.680	0.100	18165.300	1.200	5690.100
P-384	[2]	48	48	103	16.762	59.660	1.600	624.500	1.400	706.400
P-521	[3]	66	66	138	14.880	67.210	0.600	1805.500	1.100	875.200
RSA-2048	Integer Factorization	256	256	256	6618.050	0.150	1.451	689.000	0.043	23486.300
RSA-3072		384	384	384	39709.770	0.030	4.366	229.000	0.092	10843.300

Table 7. Digital Signature Algorithms. [1] ECDSA 256-bits (nistp256) , [2] ECDSA 384-bits (nistp384), [3] ECDSA 521-bits (nistp521), Units: Key sizes/Signature (bytes), Mean (ms).

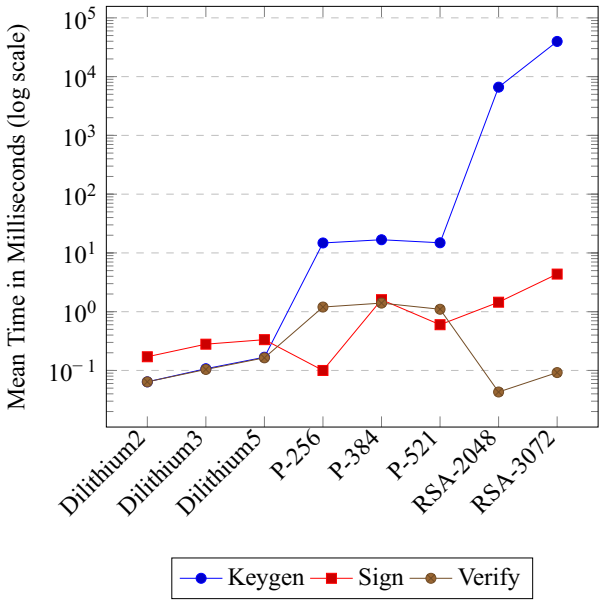

Fig. 12. Traditional vs. PQ DSA: Keygen, Sign & Verify ops.

- **Memory.** Heap usage is stable across Kyber variants (73 KB client / 82 KB server) and higher for SECP256R1 in our build (88 KB / 107 KB). Stack footprints track signature level (Fig. 16).

Performance analysis

This section interprets the benchmarks, positions them relative to prior work, and gives IIoT deployment guidance.

Comparative analysis with prior work

TLS on general-purpose CPUs. Sikeridis et al.[21] reported PQ-TLS latency increases on Intel CPUs and cloud instances. Our RPi4 results show Kyber/Dilithium can achieve single-digit to low-two-digit millisecond handshakes even on embedded-class SBCs, indicating feasibility for IIoT gateways.

Microcontroller focus. Tasopoulos et al.[38,39] emphasized performance/energy on STM32 (Cortex-M). Our work complements this by targeting a common *gateway* tier (Cortex-A72), which often terminates TLS for many field devices and must balance throughput with PQ security.

Mixed certificate deployments. Paul *et al.*[25] explored mixed classical/PQ chains in notebooks/cloud. In contrast, we quantify full PQ chains (Dilithium) and their byte/latency impact in IIoT TLS 1.3.

Lightweight three-party protocols (mobile). Works such as the anonymous quantum-safe three-party AKE for mobile[40], the post-quantum authenticated key agreement for mobile communication[41], and the more recent Ring-LWE based three-party authenticated key agreement protocol addressing vulnerabilities in prior schemes[42] highlight protocol-level constructions and simulation-driven evaluations aimed at mobile and constrained environments. Our study differs by integrating PQ primitives into *TLS 1.3 with X.509*, reporting

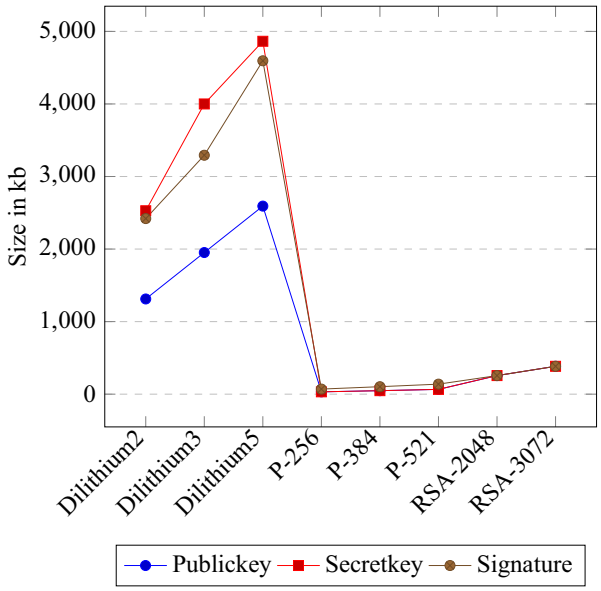

Fig. 13. Traditional vs. PQ DSA: Publickey, Secretkey & Signature Sizes.

	Key Sizes		CSR Size	Certificate Size	
Signature Algorithm	Root	EE	EE	Root	EE
Dilithium2	5.17	5.17	5.32	6.00	6.07
Dilithium3	7.96	7.96	7.32	8.00	8.07
Dilithium5	9.95	9.95	9.89	10.5	10.6
Falcon1	2.96	2.96	2.42	3.10	3.17
Falcon5	5.50	5.50	4.42	5.10	5.16
RSA1024	0.912	0.916	0.603	0.904	0.778
RSA2048	1.66	1.66	0.956	1.21	1.09
RSA3072	2.42	2.42	1.27	1.55	1.43

Table 8. Lattice-Based X.509 Certificate and key sizes (Kb).

concrete certificate sizes, handshake traces, and heap/stack/latency on IIoT-class hardware. These directions are complementary: lightweight AKEs focus on minimizing message complexity and enhancing privacy in mobile settings, whereas our results demonstrate that standardized TLS with Kyber/Dilithium is already practical for deployment at the IIoT gateway tier.

Findings

(F1) Kyber is not the bottleneck on gateways. Despite larger objects, Kyber's keygen/encap/decap latencies are low (Table 6, Fig. 10); handshake time is dominated by certificate processing and I/O framing.

(F2) Dilithium verification scales well. Verification throughput is high across Dilithium levels (Table 7, Fig. 12), suitable for server/gateway auth and client auth on capable nodes.

(F3) X.509 dominates bytes. PQ certificates inflate handshake size (Table 8, Fig. 14); short-lived EE certs, compressed chains, and gateway-assisted validation mitigate costs.

(F4) PQ can match or beat classical latency on RPi4. In our stack, Kyber512/Dilithium2–3 often achieve lower handshake time than SECP256R1 (Table 9, Fig. 17).

Deployment guidance for IIoT

Based on 6:

- **Sensors (MCU/very constrained):** Prefer gateway termination of TLS where permissible. If clients must authenticate, target *Dilithium2* and *Kyber512*; minimize chain length and use short-lived EE certs.
- **Edge gateways (SBC/RPi-class):** Use *Kyber768* + *Dilithium3* to balance assurance and throughput; enable session resumption/PSK to cap control-loop jitter.
- **Back-end/gateway anchors:** Use *Kyber1024* + *Dilithium5* to future-proof long-term trust roots and artifact signing (firmware/config).

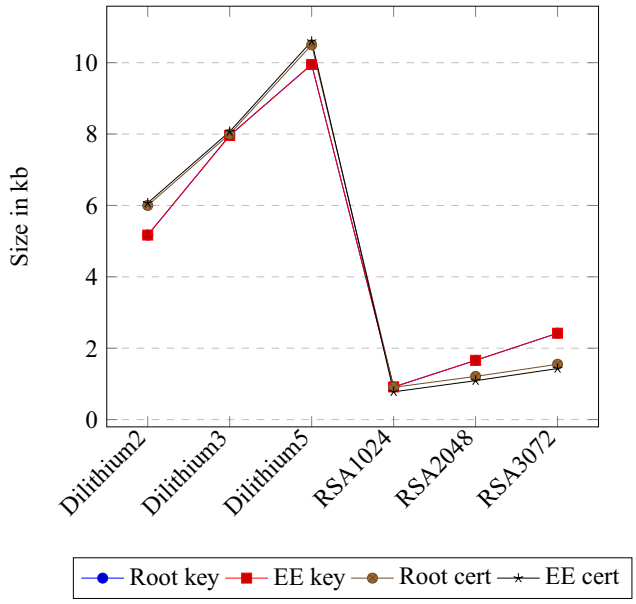

Fig. 14. Traditional vs. PQ x.509: Keys & Certificates Sizes.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	127.0.0.1	127.0.0.1	TCP	76	40070 → 11111 [SYN] Seq=0 Win=65495 Len=0 MSS=65495 SACK_PERM TSval=2250883800 TSecr=0 WS=128
2	0.000022805	127.0.0.1	127.0.0.1	TCP	76	11111 → 40070 [SYN, ACK] Seq=0 Ack=1 Win=65483 Len=0 MSS=65495 SACK_PERM TSval=2250883800 TSecr=2250883800 WS=128
3	0.000039238	127.0.0.1	127.0.0.1	TCP	68	40070 → 11111 [ACK] Seq=1 Ack=1 Win=65536 Len=0 TSval=2250883800 TSecr=2250883800
4	0.000174772	127.0.0.1	127.0.0.1	TLSv1.3	1783	Client Hello
5	0.000179200	127.0.0.1	127.0.0.1	TCP	68	11111 → 40070 [ACK] Seq=1 Ack=1716 Win=64128 Len=0 TSval=2250883800 TSecr=2250883800
6	0.007563004	127.0.0.1	127.0.0.1	TLSv1.3	1699	Server Hello
7	0.007617126	127.0.0.1	127.0.0.1	TCP	68	40070 → 11111 [ACK] Seq=1716 Ack=1632 Win=64128 Len=0 TSval=2250883808 TSecr=2250883808
8	0.007727377	127.0.0.1	127.0.0.1	TLSv1.3	96	Application Data
9	0.007734184	127.0.0.1	127.0.0.1	TCP	68	40070 → 11111 [ACK] Seq=1716 Ack=1660 Win=64896 Len=0 TSval=2250883808 TSecr=2250883808
10	0.007757801	127.0.0.1	127.0.0.1	TLSv1.3	143	Application Data
11	0.007763109	127.0.0.1	127.0.0.1	TCP	68	40070 → 11111 [ACK] Seq=1716 Ack=1735 Win=64896 Len=0 TSval=2250883808 TSecr=2250883808
12	0.008244571	127.0.0.1	127.0.0.1	TLSv1.3	6083	Application Data
13	0.008304077	127.0.0.1	127.0.0.1	TCP	68	40070 → 11111 [ACK] Seq=1716 Ack=7750 Win=61952 Len=0 TSval=2250883809 TSecr=2250883809
14	0.009578029	127.0.0.1	127.0.0.1	TLSv1.3	3391	Application Data
15	0.009618623	127.0.0.1	127.0.0.1	TCP	68	40070 → 11111 [ACK] Seq=1716 Ack=11073 Win=63360 Len=0 TSval=2250883810 TSecr=2250883810
16	0.009694198	127.0.0.1	127.0.0.1	TLSv1.3	142	Application Data
17	0.009698214	127.0.0.1	127.0.0.1	TCP	68	40070 → 11111 [ACK] Seq=1716 Ack=11147 Win=63360 Len=0 TSval=2250883810 TSecr=2250883810
18	0.010557756	127.0.0.1	127.0.0.1	TLSv1.3	8025	Application Data
19	0.013193131	127.0.0.1	127.0.0.1	TLSv1.3	4693	Application Data
20	0.013298845	127.0.0.1	127.0.0.1	TLSv1.3	142	Application Data
21	0.013396060	127.0.0.1	127.0.0.1	TLSv1.3	104	Application Data
22	0.013424437	127.0.0.1	127.0.0.1	TCP	68	11111 → 40070 [ACK] Seq=11147 Ack=14372 Win=61312 Len=0 TSval=2250883814 TSecr=2250883811
23	0.013997795	127.0.0.1	127.0.0.1	TLSv1.3	112	Application Data
24	0.014040881	127.0.0.1	127.0.0.1	TLSv1.3	92	Application Data
25	0.014059326	127.0.0.1	127.0.0.1	TCP	68	11111 → 40070 [FIN, ACK] Seq=11215 Ack=14408 Win=65536 Len=0 TSval=2250883814 TSecr=2250883814
26	0.016340751	127.0.0.1	127.0.0.1	TLSv1.3	92	Application Data
27	0.016378383	127.0.0.1	127.0.0.1	TCP	56	11111 → 40070 [RST] Seq=11216 Win=0 Len=0

Fig. 15. PQ-TLS 1.3 Handshake Packets Captured in Wireshark.

These recommendations align with the profiles introduced in Parameterization and device profiles for IIoT and our end-to-end measurements.

Threats to validity and limitations

Single platform/stack. Results are from one SBC (RPi4) and one TLS stack; different libraries or hardware cryptographic accelerators may shift absolute numbers.

No network delay/loss. Loopback testing isolates cryptographic cost but omits WAN/LAN latency, jitter, and retransmissions. Field tests with realistic links (Wi-Fi/5G/fieldbus gateways) are left to future work.

Energy not measured. Unlike microcontroller studies[39], we did not instrument power draw; this is important for battery-powered gateways and is future work.

Security analysis

A sound deployment of PQ-TLS in IIoT requires not only performance evaluation but also security validation under a clear threat model. Here, we sketch the analysis of our proposed framework against standard adversarial capabilities.

Replay attack resistance

TLS 1.3 includes nonces and ephemeral key exchanges in every handshake. When instantiated with Kyber (KEM) and Dilithium (DSA), our PQ-TLS variant inherits replay protection: encapsulated keys are bound to ephemeral randomness and session transcripts, making replayed messages invalid. Thus, prior handshake captures cannot establish valid sessions.

Authentication		Heap	Stack (kb)		Comminution (kb)		Handshake (ms)	
Client/Server	Key Exchange Algorithm	Client/Server	Client	Server	Client	Server	Client	Server
Dil3-Dil2	Kyber512	73/82	53.656	82.219	123.037	127.185	7.169	4.017
Dil5-Dil2			53.656	115.188	133.867	138.823	10.911	5.768
Dil2-Dil3			82.813	53.063	126.299	123.923	6.572	3.198
Dil5-Dil3			82.813	115.188	145.562	147.256	8.799	5.945
Dil2-Dil5			115.781	53.063	137.938	134.753	7.186	3.997
Dil3-Dil5			115.781	82.219	146.370	146.447	7.832	4.095
Dil3-Dil2	Kyber768	73/82	53.656	82.219	125.162	128.185	6.818	3.647
Dil5-Dil2			53.656	115.188	135.992	139.823	7.913	3.747
Dil2-Dil3			82.813	53.063	128.424	124.923	11.787	6.096
Dil5-Dil3			53.656	115.188	147.687	148.256	12.161	7.536
Dil2-Dil5			115.781	53.063	140.062	135.753	14.458	9.121
Dil3-Dil5			115.781	82.219	148.495	147.447	8.666	5.130
Dil3-Dil2	Kyber1024	73/82	53.656	82.219	127.600	129.497	8.202	4.510
Dil5-Dil2			53.656	115.188	138.43	141.136	13.477	12.102
Dil2-Dil3			82.813	53.063	130.861	126.235	8.445	3.788
Dil5-Dil3			82.813	115.188	150.124	149.568	8.707	4.091
Dil2-Dil5			115.781	53.063	142.500	137.065	8.083	4.563
Dil3-Dil5			115.781	82.219	150.933	148.760	12.782	8.809
Dil3-Dil2	SECP256R1	88/107	53.656	82.477	162.433	166.021	12.208	8.042
Dil5-Dil2			53.656	115.477	173.263	177.659	17.110	16.272
Dil2-Dil3			82.813	53.352	165.694	162.759	8.459	3.788
Dil5-Dil3			82.813	115.477	184.957	186.092	20.236	19.176
Dil2-Dil5			115.781	53.352	177.333	173.589	10.129	10.665
Dil3-Dil5			115.781	82.477	185.766	185.283	14.702	14.494

Table 9. Post-Quantum TLS 1.3 Handshake.

Fig. 16. Memory allocations for clients/servers: Kyber(all variants) vs P256.

Man-in-the-Middle (MITM) resistance

Under the Canetti–Krawczyk (CK) model for authenticated key exchange, PQ-TLS inherits forward secrecy and channel authentication. In our scheme, authenticity is provided by Dilithium-signed X.509 certificates; secrecy is guaranteed by Kyber KEM. A MITM adversary cannot generate valid certificate chains or complete KEM decapsulation without the legitimate private key, ensuring confidentiality and integrity against active interception.

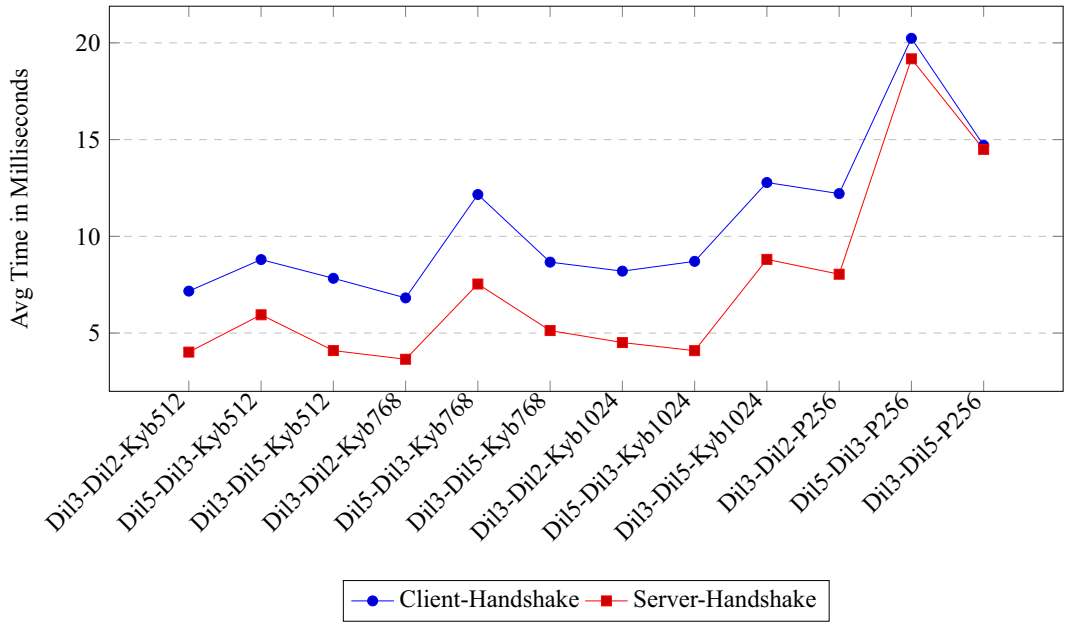

Fig. 17. Authentication/Key-Exchange Algorithm combinations comparing client/server handshake.

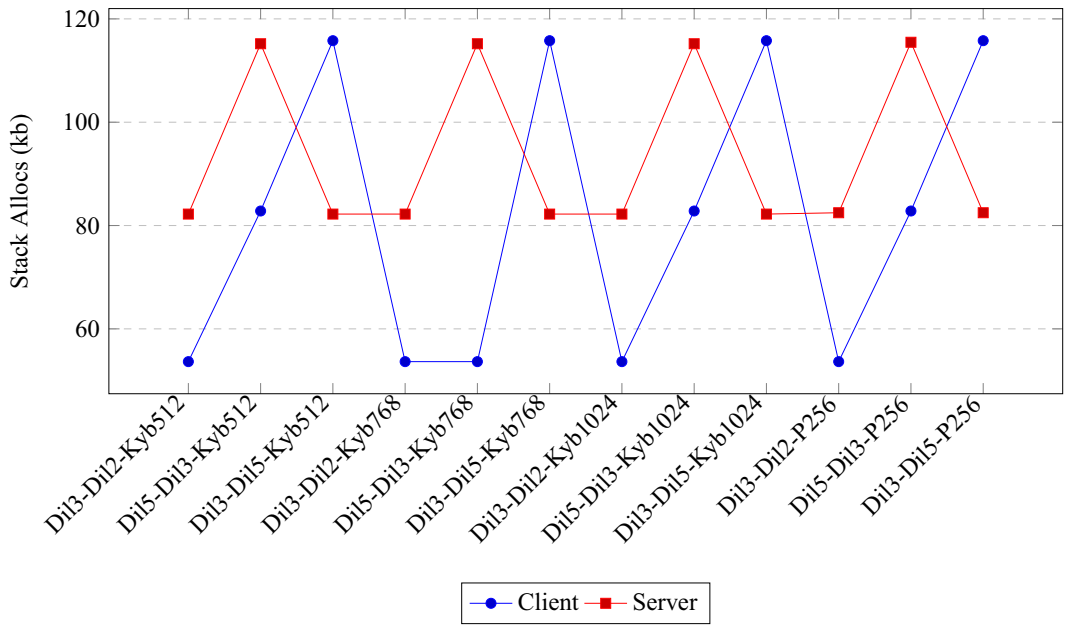

Fig. 18. Authentication/Key-Exchange Algorithm combinations comparing memory stack allocations.

Forward secrecy and key compromise

By relying on ephemeral KEM key pairs (Kyber), session keys are independent of long-term keys. Even if a device's Dilithium private key is later compromised, past sessions remain secure. This property is critical for long-lived IIoT nodes vulnerable to eventual key extraction.

Side-channel considerations

Resource-constrained IIoT devices are especially prone to timing, power, and electromagnetic side-channel attacks. Our experiments on Raspberry Pi do not capture these risks. However, lattice-based primitives like Kyber and Dilithium are under active hardening for side-channel resistance, with masked and constant-time implementations. We acknowledge this as a limitation and an area for future hardware-level evaluation.

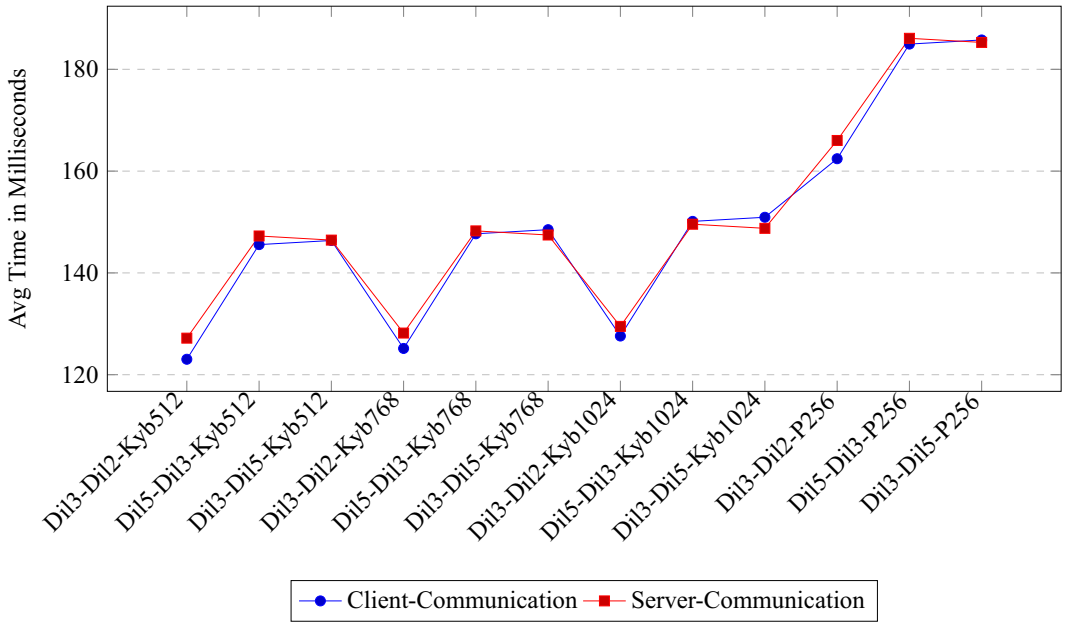

Fig. 19. Authentication/Key-Exchange Algorithm combinations comparing client/server communication kb used.

Summary

Our analysis shows that PQ-TLS 1.3 with Kyber and Dilithium maintains resistance to replay and MITM attacks under the CK model, provides forward secrecy, and inherits TLS transcript binding. Side-channel resistance remains an open practical challenge, but the integration of hardened libraries (e.g., liboqs with masked implementations) offers a mitigation path.

Discussion

Deployment challenges

While Raspberry Pi benchmarks show PQ-TLS feasibility, actual IIoT deployments face constraints:

- **Memory and storage limits:** PQ X.509 certificates (5–10 KB) strain devices with 256 KB flash.
- **Backward compatibility:** IIoT deployments often require hybrid certificates (RSA/ECDSA + Dilithium) to interoperate with legacy systems.
- **Update lifecycles:** Long-lived IIoT devices may lack secure update channels, complicating cryptographic agility.

Standardization landscape

Our scheme aligns with active standardization efforts. IETF drafts on PQ-TLS, NIST's post-quantum competition finalization, and the U.S. CNSA 2.0 roadmap all mandate lattice-based algorithms (Kyber, Dilithium) for near-term adoption. PQ X.509 certificate encoding, though still experimental, is under evaluation by OQS–OpenSSL and ETSI technical groups, suggesting that interoperability testing will soon be possible.

Hardware feasibility

We selected Raspberry Pi 4 as a baseline gateway device, representing mid-tier IIoT controllers. Results show PQ handshakes under 15 ms and stable memory use (73–82 KB heap). However, microcontrollers (e.g., Cortex-M, 1 MB RAM) present steeper challenges. While Kyber-512 may be feasible with lightweight libraries, Dilithium signatures and full PQ certificates remain prohibitive without hardware acceleration or FPGA offloading. These constraints highlight the importance of tiered deployment strategies: PQ-TLS at gateways, lightweight AKEs at edge sensors.

Sector-specific recommendations

Beyond generic deployment, our benchmarks suggest tailored algorithm profiles for different sectors:

Energy & Utilities: Dilithium-5 + Kyber-1024 for highest assurance.
Oil & Gas: Dilithium-3/5 + Kyber-768 for balanced latency/security.
Mining & Agriculture: Dilithium-2 + Kyber-512 for reduced overhead.
Smart Cities & Manufacturing: Dilithium-3/5 + Kyber-768 for scalability.
Healthcare: Dilithium-3/2 + Kyber-1024 for strong privacy guarantees.
This mapping demonstrates how PQ-TLS parameterization can align with sectoral risk and resource profiles.

Conclusion

This work demonstrates the practical feasibility of deploying TLS 1.3 secured with lattice-based post-quantum algorithms in Industrial IoT (IIoT) environments. The study integrates NIST-standardized CRYSTALS-Kyber (KEM) and CRYSTALS-Dilithium (DSA) within TLS 1.3 and X.509 certificates, providing a concrete implementation and evaluation on IIoT-class hardware.

Benchmarking on a Raspberry Pi 4 platform shows that PQ-TLS 1.3 maintains handshake latency below 15 ms and memory usage under 100 KB, with the primary overhead attributed to certificate size rather than cryptographic computation. The results confirm that Kyber-512 and Dilithium-2 are suitable for low-power sensors and edge controllers, while Kyber-1024 and Dilithium-5 provide stronger security for gateways with minimal latency impact. Furthermore, the paper offers sector-specific recommendations that align algorithm parameters with IIoT industry requirements.

The main limitations of this work include the use of a single hardware platform, controlled network conditions, and the absence of microcontroller-level and energy-consumption evaluations. These constraints will be addressed in future work through field testing on heterogeneous IIoT networks, hardware acceleration on FPGAs and embedded systems, and exploration of hybrid migration strategies combining classical and post-quantum primitives.

In summary, PQ-TLS 1.3 using Kyber and Dilithium provides a viable pathway toward quantum-resistant authentication and key establishment in IIoT gateways. This study bridges theory and implementation, contributing a reproducible foundation for the secure migration of industrial infrastructures to the post-quantum era.

Data availability

This study is based on the implementation and evaluation of post-quantum cryptographic protocols using publicly available open-source libraries. The X.509 certificate generation was performed using liboqs version 0.8.0 from the Open Quantum Safe project, while the TLS 1.3 integration utilized the Open Quantum Safe fork of WolfSSL version 5.6.3. As no experimental or proprietary datasets were generated or analyzed, there is no additional data associated with this work.

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References

1. National Academies of Sciences, E., Medicine et al. *Quantum computing: progress and prospects*. (2019).
2. Shor, P. W. Introduction to quantum algorithms. *Proc. Symp. Appl. Math.* **58**, 143–160 (2002).
3. Post-quantum cryptography CSRC CSRC - csrc.nist.gov. https://csrc.nist.gov/projects/post-quantum-cryptography.
4. https://csrc.nist.gov/News/2016/Public-Key-Post-Quantum-Cryptographic-Algorithms.
5. Federal Register : Request Access - federalregister.gov. https://www.federalregister.gov/documents/2016/12/20/2016-30615/announcing-request-for-nominations-for-public-key-post-quantum-cryptographic-algorithms.
6. NCCoE Migration to PQC Community of Interest Meeting NCCoE -nccoe.nist.gov. https://www.nccoe.nist.gov/get-involved/attend-events/nccoe-migration-pqc-community-interest-meeting/post-webinar-materials (2023).
7. NCCoE Migration to PQC Community of Interest Meeting NCCoE - nccoe.nist.gov. https://www.nccoe.nist.gov/get-involved/attend-events/nccoe-migration-pqc-community-interest-meeting (2023).
8. https://www.cisa.gov/news-events/news/cisa-nsa-and-nist-publish-new-resource-migrating-post-quantum-cryptography (2023).
9. Three Draft FIPS for Post-Quantum Cryptography CSRC - csrc.nist.gov. https://csrc.nist.gov/news/2023/three-draft-fips-for-post-quantum-cryptography (2023).
10. Post-Quantum Cybersecurity Resources - nsa.gov. https://www.nsa.gov/Cybersecurity/Post-Quantum-Cybersecurity-Resources/.
11. Tawfeeq, N. H., Yousif, M., Al-Shareeda, M. A., Almaiah, M. & Shehab, R. Lightweight and quantum-resistant authentication for the Internet of Drones (IoD) using Dilithium signatures. *Int. J. Innov. Res. Sci. Stud.* https://doi.org/10.53894/ijirss.v8i2.5825 **8**(2), 2842-2853 (2020).
12. Samandari, J. & Gritti, C. Post-Quantum Authentication and Integrity in 3-Layer IoT Architectures. *Conference on Privacy, Security and Trust* https://doi.org/10.1109/PST62714.2024.10788057 (2024).
13. Samandari, J. & Gritti, C. Post-Quantum Authentication in the MQTT Protocol. *J. Cybersecur. Priv.* https://doi.org/10.3390/jcp3030021 (2023).
14. Prajwal, C. P., Mohan, A. S., Reddy, D. N. & Nimmy, K. Quantum-Safe Authentication Protocol leveraging qPUF for Industrial Internet of Things. *Int. Conf. on Comput. Commun. Netw. Technol.* https://doi.org/10.1109/ICCCNT61001.2024.10725580 (2024).
15. Kumari, S., Singh, M., Singh, R. & Tewari, H. A post-quantum lattice based lightweight authentication and code-based hybrid encryption scheme for IoT devices. *Comput. Networks* https://doi.org/10.1016/j.comnet.2022.109327 (2022).
16. Gupta, D. S. Pilike: Post-Quantum identity-based lightweight authenticated key exchange protocol for IIoT environments. *IEEE Syst. J.* https://doi.org/10.1109/JSYST.2023.3335217 (2024).
17. McLoughlin, C., Gritti, C. & Samandari, J. Full Post-Quantum Datagram TLS Handshake in the Internet of Things. *International Conference on Codes, Cryptology, and Information Security* https://doi.org/10.1007/978-3-031-33017-9_4 (2023).
18. Schöffel, M., Lauer, F., Rheinländer, C. & Wehn, N. Secure IoT in the Era of Quantum computers-where are the bottlenecks? *Italian Natl. Conf. on Sensors* https://doi.org/10.3390/s22072484 (2022).
19. Banegas, G., Zandberg, K., Herrmann, A., Baccelli, E. & Smith, B. A. Quantum-Resistant Security for Software Updates on Low-power Networked Embedded Devices. IACR Cryptology ePrint Archive (2021).
20. Behnia, R. & Yavuz, A. Towards practical post-quantum signatures for resource-limited Internet of Things. *Asia-PacificComput. Syst. Archit. Conf.* https://doi.org/10.1145/3485832.3488023 (2021).
21. Sikeridis, D., Kampanakis, P. & Devetsikiotis, M. Post-quantum authentication in tls 1.3: a performance study. *Cryptology ePrint Archive* (2020).
22. Dharminder, D. & Mishra, D. Lcppa: Lattice-based conditional privacy preserving authentication in vehicular communication. *Trans. Emerg. Telecommun. Technol.* **31**, e3810 (2020).
23. Paul, S. & Guerin, E. Hybrid opc ua: enabling post-quantum security for the industrial internet of things. In *2020 25th IEEE International Conference on Emerging Technologies and Factory Automation (ETFA)*, vol. 1, 238–245 (IEEE, 2020).

24. Ravi, P., Sundar, V. K., Chattopadhyay, A., Bhasin, S. & Easwaran, A. Authentication protocol for secure automotive systems: Benchmarking post-quantum cryptography. In *2020 IEEE International Symposium on Circuits and Systems (ISCAS)* 1–5 (IEEE, 2020).
25. Paul, S., Kuzovkova, Y., Lahr, N. & Niederhagen, R. Mixed certificate chains for the transition to post-quantum authentication in tls 1.3. In *Proceedings of the 2022 ACM on Asia Conference on Computer and Communications Security* 727–740 (2022).
26. Wang, F., Wang, J. & Yang, W. Efficient incremental authentication for the updated data in fog computing. *Future Gener. Comput. Syst.* **114**, 130–137 (2021).
27. Tang, Y., Ba, Y., Li, L., Wang, X. & Yan, X. Lattice-based public-key encryption with conjunctive keyword search in multi-user setting for iiot. *Clut. Comput.* **25**, 2305–2316 (2022).
28. Liu, J. et al. A post quantum secure multi-party collaborative signature with deterability in the industrial internet of things. *Future Gener. Comput. Syst.* **141**, 663–676 (2023).
29. Lohachab, A. et al. Using quantum key distribution and ecc for secure inter-device authentication and communication in iot infrastructure. In *Proceedings of 3rd International Conference on Internet of Things and Connected Technologies (ICIoTCT)* 26–27 (2018).
30. Latif, M. K., Jacinto, H. S., Daoud, L. & Rafla, N. Optimization of a quantum-secure sponge-based hash message authentication protocol. In *2018 IEEE 61st International Midwest Symposium on Circuits and Systems (MWSCAS)* 984–987 (IEEE, 2018).
31. Crockett, E., Paquin, C. & Stebila, D. Prototyping post-quantum and hybrid key exchange and authentication in tls and ssh. *Cryptology ePrint Archive* (2019).
32. Chikouche, N., Cayrel, P.-L., Mboup, E. H. M. & Boidje, B. O. A privacy-preserving code-based authentication protocol for internet of things. *J. Supercomput.* **75**, 8231–8261 (2019).
33. Fujioka, A., Takashima, K., Terada, S. & Yoneyama, K. Supersingular isogeny diffie-hellman authenticated key exchange. In *Information Security and Cryptology-ICISC 2018: 21st International Conference, Seoul, South Korea*, November 28–30, 2018, Revised Selected Papers 21, 177–195 (Springer, 2019).
34. Zhang, X., Xu, C., Wang, H., Zhang, Y. & Wang, S. Fs-peks: Lattice-based forward secure public-key encryption with keyword search for cloud-assisted industrial internet of things. *IEEE Trans. Dependable Secure Comput.* **18**, 1019–1032 (2019).
35. Round 1 Submissions - Post-Quantum Cryptography CSRC CSRC - csrc.nist.gov. https://csrc.nist.gov/Projects/post-quantum-cryptography/post-quantum-cryptography-standardization/round-1-submissions.
36. Bos, J. et al. Crystals-kyber: a cca-secure module-lattice-based kem. In *2018 IEEE European Symposium on Security and Privacy (EuroS &P)* 353–367 (IEEE, 2018).
37. Ducas, L. et al. *Crystals-dilithium: Digital signatures from module lattices*. (2018).
38. Tasopoulos, G. et al. Performance evaluation of post-quantum tls 1.3 on resource-constrained embedded systems. In *International Conference on Information Security Practice and Experience* 432–451 (Springer, 2022).
39. Tasopoulos, G. et al. Energy consumption evaluation of post-quantum tls 1.3 for resource-constrained embedded devices. *Cryptology ePrint Archive* (2023).
40. Chaudhary, D. et al. Anonymous quantum safe construction of three party authentication and key agreement protocol for mobile devices. *IEEE Access* **12**, 74572–74585 (2024).
41. Kumar, U., Garg, M., Kumari, S. & Dharminder, D. A construction of post quantum secure and signal leakage resistant authenticated key agreement protocol for mobile communication. *Trans. Emerg. Telecommun. Technol.* **34**, e4695 (2022).
42. Kumar, U., Garg, M. & Chaudhary, D. Design and analysis of a post-quantum secure three party authenticated key agreement protocol based on ring learning with error for mobile device. *J. Supercomput.* **81**, 9 (2024).

Author contributions

- Abdul Basit Shahid and Khwaja Mansoor were primarily responsible for drafting the main body of the manuscript, including the formulation of the core content, background, and literature review. - Yawar Abbas Bangash and Waseem Iqbal contributed significantly to the design and development of the proposed model and methodology. In addition to this, Yawar Abbas Bangash also oversaw the overall project management, coordination among team members, and integration of contributions across sections. - Shynar Mussiraliyeva provided critical feedback, performed thorough proofreading, and actively supported the structural revisions and improvements throughout the manuscript. - All authors reviewed and approved the final version of the manuscript and agree to be accountable for all aspects of the work.

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Declarations

Competing interests

The authors decalre no competing interests.

Additional information

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